Ecole Supérieure Privée
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Integration Project Report

Project Title:

Intelligent platform for managing training programs and professional certifications

SkillHub Pro

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General Introduction

In the digital era, continuous learning and skill development have become essential to meet the demands of a constantly evolving labor market. Both organizations and individuals face the need to acquire new skills to adapt to technological, economic, and social transformations.

In this context, training platforms play a central role in providing access to knowledge and enhancing professional competencies. However, traditional training management systems often have limitations in terms of personalization, efficiency, and intelligent data utilization. Learners sometimes struggle to identify the most suitable courses for their needs, while organizations face challenges in managing and monitoring training programs.

To address these issues, this integration project aims to design and develop an intelligent platform for managing professional training and certifications. This platform will centralize the management of courses, users, enrollments, and assessments while integrating an intelligent recommendation module based on Machine Learning technques. The goal is to provide a more efficient, personalized, and scalable digital environment that meets the expectations of both learners and organizations.

This report presents the various stages of project design and planning, focusing on needs analysis, system modeling, adopted architecture, and development planning. It thus provides a solid foundation for the technical implementation of the platform.

Chapter 1

Project Framework Presentation

Introduction

This chapter aims to present the general framework of the integration project. It provides the context in which the project is carried out, outlines the problem it addresses, and describes the work to be completed. A review of existing solutions is also presented to identify their limitations and justify the need for the proposed solution. Finally, this chapter introduces the chosen methodology and the modeling formalism used for system design.

1.1 General framework of the work

The digitalization of training and certification processes has become essential to meet the growing demand for professional skills. Digital platforms now make it easier to manage training programs and improve learner support.

In this context, an intelligent platform for managing training and certifications is proposed to optimize the organization of training pathways and provide recommendations tailored to users' profiles. The platform will also include a recommendation module based on Machine Learning to suggest the most relevant courses. It aims to offer a scalable and efficient solution that meets the needs of both learners and organizations.

1.1.1 Problem Statement

Learners often struggle to find training programs that match their skills, while organizations face challenges in effectively monitoring training pathways and certifications. Therefore, it is necessary to provide a solution capable of managing training programs while automatically recommending the most relevant courses. This recommendation will be personalized based on each learner's profile and skills.

1.1.2 Description of the Work to Be Done

The project involves designing an intelligent platform to manage training programs, users, enrollments, and certifications. The system will also include a recommendation module to suggest the most suitable courses for each learner. The minimum viable product (MVP) will cover user management, course management, enrollments, and basic reporting, with the recommendation module as a core feature.

1.2 Study of Existing Solutions

1.2.1 Description of the Existing Situation

Currently, several platforms are used for managing training programs and professional certifications. The most well-known are Moodle, Coursera, and Udemy, each with its own characteristics.

Moodle: a widely used open-source platform in educational institutions and companies. It allows the creation of online courses, user management, assessments, and discussion forums.
Coursera: a commercial platform offering online courses and certificates issued by partner universities or institutions. It provides an intuitive interface and a large course catalog.
Udemy: a commercial platform aimed at the general public, with a diverse catalog and features for tracking progress and issuing certificates.

Main Features

- User and enrollment management
- Course publication and access
- Assessment tracking and certificate issuance
- Access to online learning resources

Technologies Used

- **Moodle**: PHP, MySQL, JavaScript
- **Coursera and Udemy**: primarily modern web architectures (Java, Python, web frameworks), relational databases, and basic recommendation systems for some content.

This review highlights the functionalities offered by current platforms while setting the stage for discussing their limitations and the need for a more intelligent and adaptable solution.

To illustrate the current state of such systems, examples of existing training platforms are presented in Appendix A.

1.2.2 Critique of the Existing Situation

Although existing platforms such as Moodle, Coursera and Udemy provide useful features for managing training programs and certifications, they present several limitations that reduce their effectiveness and adaptability to the current needs of learners and organizations.

Table 1.1: Main limitations of existing training platforms

Platform	**Main limitations**
Moodle	• Complex user interface for non-technical users • Limited personalization of learning paths • Lack of intelligent recommendation mechanisms
Coursera	• Rigid and predefined learning paths • Limited adaptation to individual learner profiles • Restricted pedagogical customization options
Udemy	• Weak pedagogical supervision and learner monitoring • Limited assessment and certification structure • Absence of formalized learning pathways

These limitations justify the need for a more intelligent and adaptive training management solution, which is presented in the following sections.

1.2.3 Proposed Solution

To address the identified shortcomings, the proposed solution is to develop an intelligent platform *SkillHub Pro* for managing training and professional certifications that is adapted to the needs of learners and organizations.

Key Components of the Solution

- **Intelligent course recommendation**: a module based on machine learning techniques will automatically suggest the most relevant courses according to each user's profile, skills, and goals.
- **Personalized progress tracking**: training managers will be able to monitor learner progress in near-real-time and adapt learning paths to specific needs.
- **Intuitive and ergonomic interface**: the platform will be designed for ease of use by learners, instructors and administrators alike.
- **Advanced reporting tools**: dashboards and detailed reports will help analyze performance, identify weaknesses and improve course quality.
- **Flexibility and scalability**: the platform will support the integration of new features and adapt to various organizational contexts and types of training.

Added Value Compared to Existing Platforms

- **Personalized learning paths**: unlike current solutions, each learner will receive recommendations tailored to their needs and competencies.
- **Improved pedagogical monitoring**: instructors and managers will be able to steer training more effectively using detailed analytics.
- **Better user experience**: a clear and intuitive interface will increase learner engagement and success.
- **Data-driven decisions**: managers will have reliable data to continuously refine content and training management.

In summary, the proposed solution offers an intelligent, personalized and user-centered approach that addresses the shortcomings of existing platforms and brings significant added value to professional training.

1.3 Methodological Choices and Adopted Formalism

1.3.1 Comparative Study between Classical and Agile Methods

Classical methods, such as the Waterfall model, follow a linear and sequential process where each phase must be completed before moving to the next one. This lack of flexibility makes

them less suitable for projects in which requirements may evolve over time

In contrast, Agile approaches rely on iterative and incremental cycles that promote continuous feedback, close collaboration with stakeholders, and frequent delivery of functional product increments.

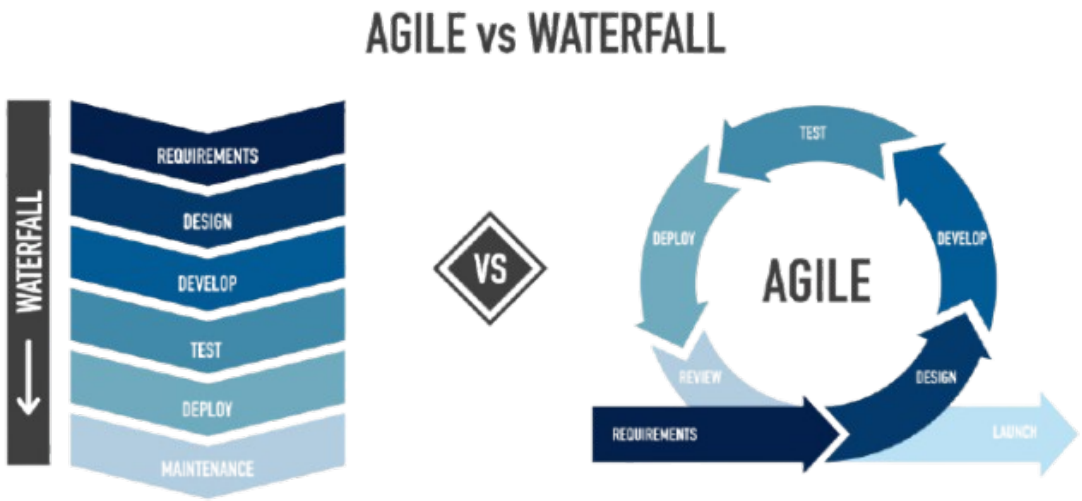

Figure 1.1: Comparison between classical and agile methods

1.3.1.1 Why Scrum?

Scrum was chosen for its ability to manage complex and evolving projects through short development cycles (sprints) and regular reviews with stakeholders. This approach improves alignment with expectations, facilitates early feedback, and reduces the risk associated with changing requirements.

1.3.1.2 Scrum Framework

The Scrum framework organizes work into fixed-length iterations called *sprints* (we propose 2-week sprints). Each sprint delivers a potentially shippable increment. Key Scrum artifacts include the Product Backlog, the Sprint Backlog and the Definition of Done. Ceremonies comprise Sprint Planning, Daily Scrum, Sprint Review and Sprint Retrospective. These practices promote transparency, inspection and adaptation throughout the development process.

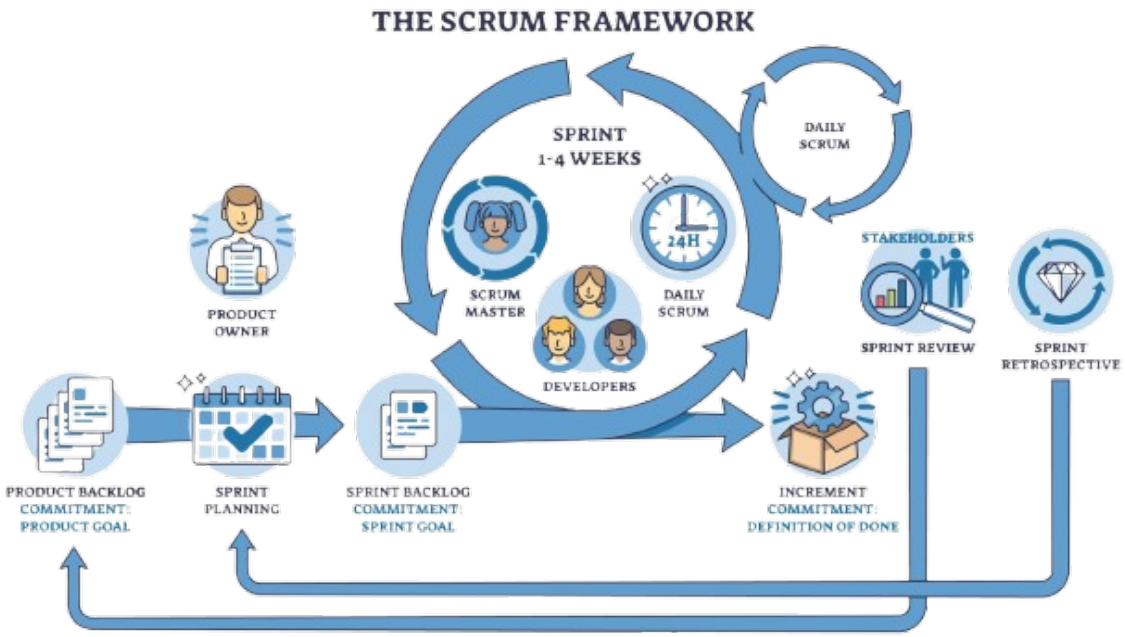

Figure 1.2: Scrum framework: roles, artifacts and ceremonies

1.3.1.3 The Scrum Team

- **Product Owner**: responsible for defining priorities and maintaining the product backlog.
- **Development Team**: implements features and ensures the quality of deliverables.
- **Scrum Master**: facilitates the Scrum process, removes impediments and protects the team from external disruptions.

For this project, roles will be assigned among the six team members. A suggested distribution is: 1 Product Owner, 1 Scrum Master , 3 Developers (backend/frontend), and 1 Data Scientist / QA engineer. The final assignment should be confirmed by the team.

1.3.2 Adopted Formalism

To model the platform, we will use UML, which allows clear representation of requirements and design through multiple diagram types. The report will include:

- Use case diagrams,
- Class diagrams,
- Sequence diagrams,

Conclusion

In summary, Scrum and UML have been selected to support iterative delivery and clear system modeling. Chapter 2 will present the detailed architecture, requirements analysis and some prototypes.

Chapter 2

Planning and System Architecture

Introduction

This chapter presents the planning and system architecture of the SkillHub Pro platform. It identifies the main stakeholders and analyzes both functional and non-functional requirements. The chapter also describes the logical and physical architecture of the system based on the MVC and n-tier models, and introduces UML modeling as well as low-fidelity prototypes that guide the design and future implementation of the platform.

2.1 Stakeholders and Actors Identification

The SkillHub Pro platform involves several stakeholders and actors who interact with the system according to their roles and responsibilities. Identifying these actors helps define system boundaries and functional requirements.

- **Visitor**: a non-authenticated user who can consult the course catalog and register on the platform.
- **Learner**: an authenticated user who can enroll in courses, manage their profile, follow learning progress, take assessments, download certificates, receive notifications, and obtain personalized course recommendations.
- **Instructor**: responsible for creating and managing course content, managing assessments, and monitoring learners' progress through analytics dashboards.
- **Administrator**: responsible for managing users (learners and instructors), roles and permissions, and accessing global reports for platform supervision.
- **Recommendation Engine**: a system component that generates personalized course recommendations based on learners' profiles, skills, and learning history.

2.2 Requirements Analysis

2.2.1 Functional Requirements

Functional requirements describe the services and functionalities that the system must provide to its users. For the intelligent training and certification management platform, these requirements define how different actors interact with the system and how core processes such as training management, evaluation, certification, and recommendation are handled.

The functional requirements of the *SkillHub Pro* platform are organized according to the main system features and are described below.

FR1 – User Management The system shall allow users to register, authenticate, and manage their personal profiles. It shall support multiple user roles, including learners, instructors, training managers, and administrators, each with specific access rights and permissions.

FR2 – Training Catalog Management The system shall provide a dynamic training catalog enabling users to browse, search, and filter available courses. Instructors and administrators shall be able to create, update, and delete training content.

FR3 – Enrollment Management The system shall allow learners to enroll in training programs and manage their enrollments. It shall track enrollment status and learner participation throughout the learning process.

FR4 – Learning Progress Tracking The system shall monitor learners' progress by recording course completion, assessment results, and learning milestones. Training managers and instructors shall be able to consult progress indicators through dashboards and reports.

FR5 – Assessment and Evaluation Management The system shall support the creation and management of assessments, including quizzes, assignments, and exams. It shall automatically record evaluation results and make them available to learners and instructors.

FR6 – Certification Management The system shall generate and deliver digital certificates upon successful completion of training programs. Certificates shall be securely stored and made available for download and verification.

FR7 – Intelligent Course Recommendation The system shall include a machine learning–based recommendation module that automatically suggests relevant training courses to learners.

Recommendations shall be generated based on learners' profiles, skills, learning history, and preferences.

FR8 – Reporting and Analytics The system shall provide reporting and analytics tools to training managers and administrators. These tools shall enable the analysis of learner performance, training effectiveness, and overall platform usage.

Table 2.1: Functional Requirements of the SkillHub Pro Platform

ID	Functional Requirement
FR1	User registration, authentication, and role management
FR2	Training catalog creation, update, and consultation
FR3	Enrollment and participation management
FR4	Learning progress tracking and monitoring
FR5	Assessment and evaluation management
FR6	Digital certification generation and delivery
FR7	Intelligent course recommendation using machine learning
FR8	Reporting and analytics for training supervision

2.2.2 Non-Functional Requirements

Non-functional requirements describe the quality attributes and constraints that the system must satisfy in order to deliver a reliable, secure, and efficient user experience. Unlike functional requirements, they focus on system behavior, performance, and operational characteristics. For an intelligent training and certification management platform, these requirements are essential to ensure scalability, security, and long-term sustainability.

Performance The platform shall provide fast and responsive interactions for all users. Key operations such as browsing the training catalog, enrolling in courses, accessing learning resources, and receiving recommendations shall be executed with minimal latency, even during peak usage periods.

Scalability The system shall be designed to support growth in terms of users, training content, and data volume. It shall support horizontal scalability, allowing additional service instances to be deployed without impacting system stability or performance.

Availability and Reliability The platform shall ensure high availability and continuous access to services. Reliability mechanisms shall be implemented to handle failures gracefully and to maintain system functionality in the event of partial outages.

Security Security is a critical requirement for the platform. Access to the system shall be protected through robust authentication and authorization mechanisms. Sensitive information, including personal user data and certification records, shall be securely stored and transmitted using encryption and secure communication protocols.

Data Privacy and Compliance The platform shall comply with data protection regulations such as the General Data Protection Regulation (GDPR). Personal data shall be collected and processed transparently, ensuring confidentiality and respecting user consent throughout the learning lifecycle.

Usability and Accessibility The system shall provide an intuitive and user-friendly interface adapted to different user profiles. The platform shall be accessible from various devices, including desktops, tablets, and smartphones, ensuring a consistent learning experience.

Maintainability The architecture of the platform shall be modular and well-documented to facilitate maintenance and future enhancements. This approach enables the integration of new features or updates without disrupting the overall system.

Interoperability The platform shall support integration with external systems, such as human resources management tools or third-party learning platforms, through standardized and well-documented application programming interfaces (APIs).

Table 2.2: Non-Functional Requirements of the SkillHub Pro Platform

Category	**Description**
Performance	Fast response time and efficient execution of core operations
Scalability	Ability to handle system growth and increased workloads
Availability	Continuous and reliable access to platform services
Security	Protection of data and controlled access to the system
Data Privacy	Compliance with regulations and respect of user consent
Usability	Intuitive and accessible user interface across devices
Maintainability	Ease of maintenance, evolution, and feature extension
Interoperability	Integration with external systems via standardized APIs

2.3 System Architecture

The architecture of SkillHub Pro is designed to ensure clear separation of concerns, scalability, and maintainability. The system follows a Model–View–Controller (MVC) pattern combined with an n-tier physical deployment.

2.3.1 Architecture Overview

SkillHub Pro is composed of three main layers:

- Angular frontend, representing the **View** layer,
- Spring Boot backend, implementing the **Controller and Model** layers,
- A relational database for data persistence.

Communication between the frontend and backend is based on RESTful APIs and JSON messages.

2.3.2 Logical Architecture

The logical architecture follows the MVC pattern implemented in Spring Boot.

2.3.2.1 Logical Components Description

The main logical components are:

- **View (Angular):** user interface of the platform.
- **Controllers (Spring Boot):** handle HTTP requests from the frontend.
- **Services:** implement business logic.
- **Repositories:** manage database access using JPA.
- **Entities:** represent the data model mapped to database tables.

2.3.2.2 Logical Architecture Diagram

Figure 2.1 illustrates the logical MVC architecture of SkillHub Pro, where Angular communicates with Spring Boot, which interacts with the database through JPA.

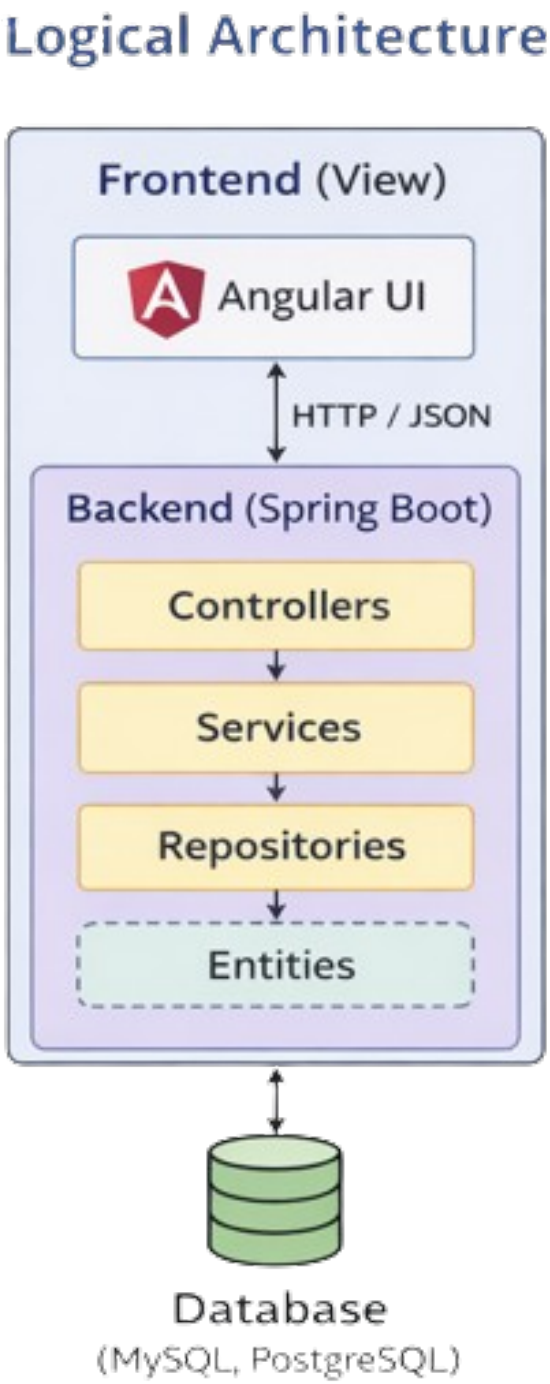

Figure 2.1: Logical Architecture of SkillHub Pro

2.3.3 Physical Architecture

The physical architecture describes how the system components are deployed in a real environment.

2.3.3.1 n-Tier Physical Architecture

The system follows an n-tier architecture implemented as four tiers:

- **Tier 1 – Client:** web browser,
- **Tier 2 – Frontend:** Angular application (Node.js, port 4200),
- **Tier 3 – Backend:** Spring Boot application (Tomcat, port 8080),
- **Tier 4 – Database:** MySQL (3306) or PostgreSQL (5432).

2.3.3.2 Physical Architecture Diagram

Figure 2.2 shows the physical deployment: the client accesses Angular, which sends REST requests to Spring Boot, which communicates with the database.

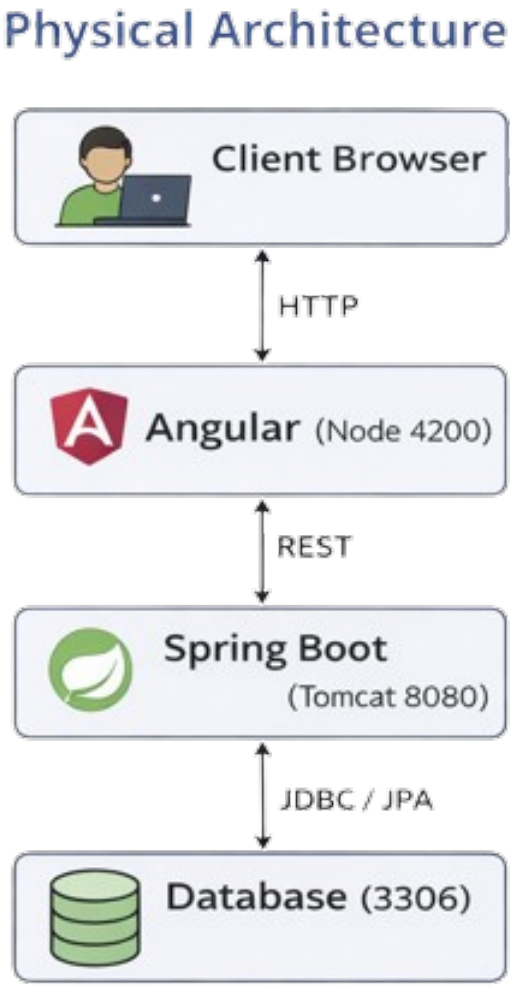

Figure 2.2: Physical Deployment Architecture of SkillHub Pro

2.4 System Modeling (UML)

2.4.1 Use Case Diagram

A high-level use case diagram will show interactions between actors and main use cases .

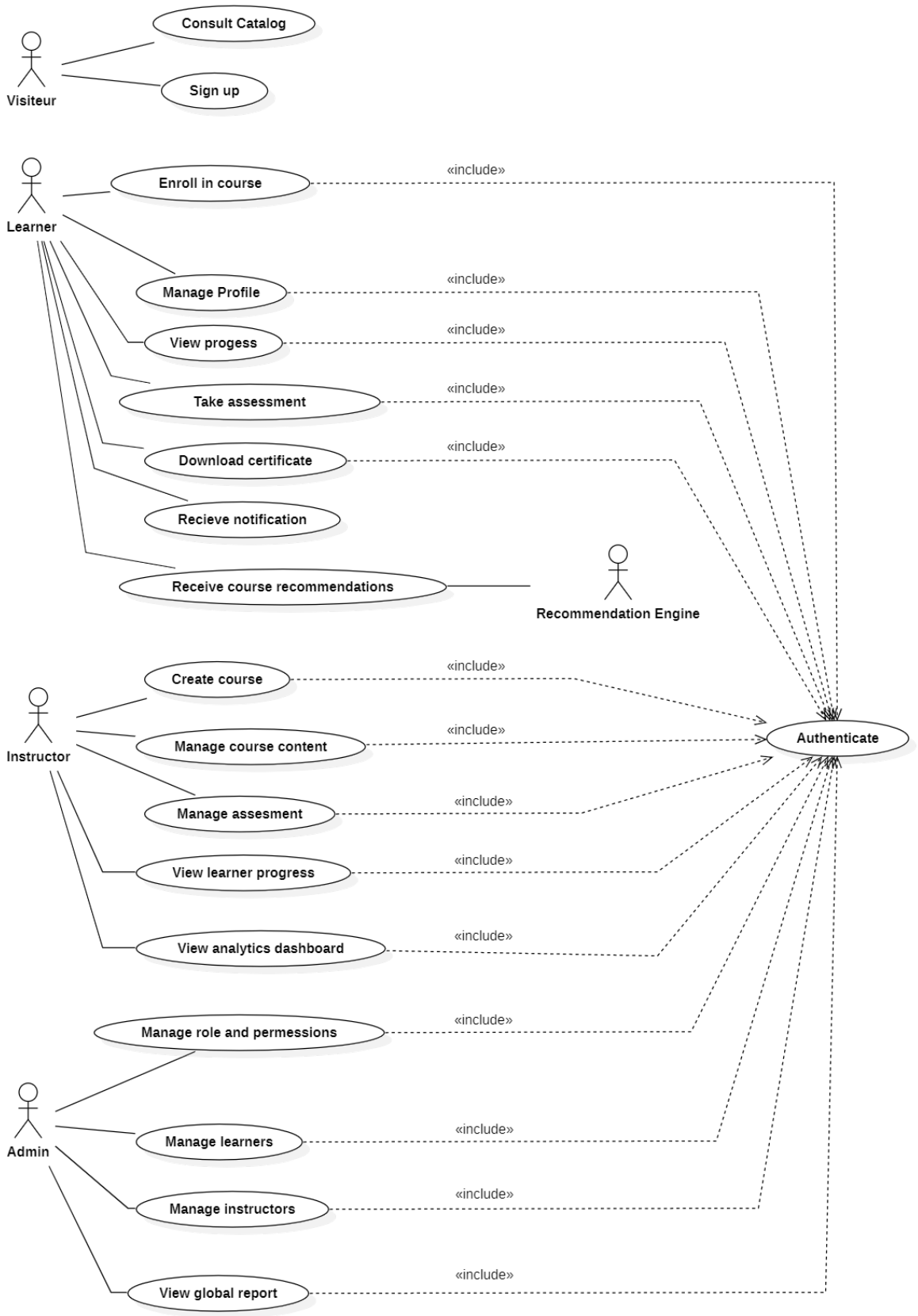

Figure 2.3: Global Use Case Diagram of SkillHub Pro

2.4.2 Class Diagram

Define domain classes: `User`, `Profile`, `Course`, `Module`, `Enrollment`, `Assessment`, `Certificate`, `Recommendation`, `Skill`, with relationships and key attributes.

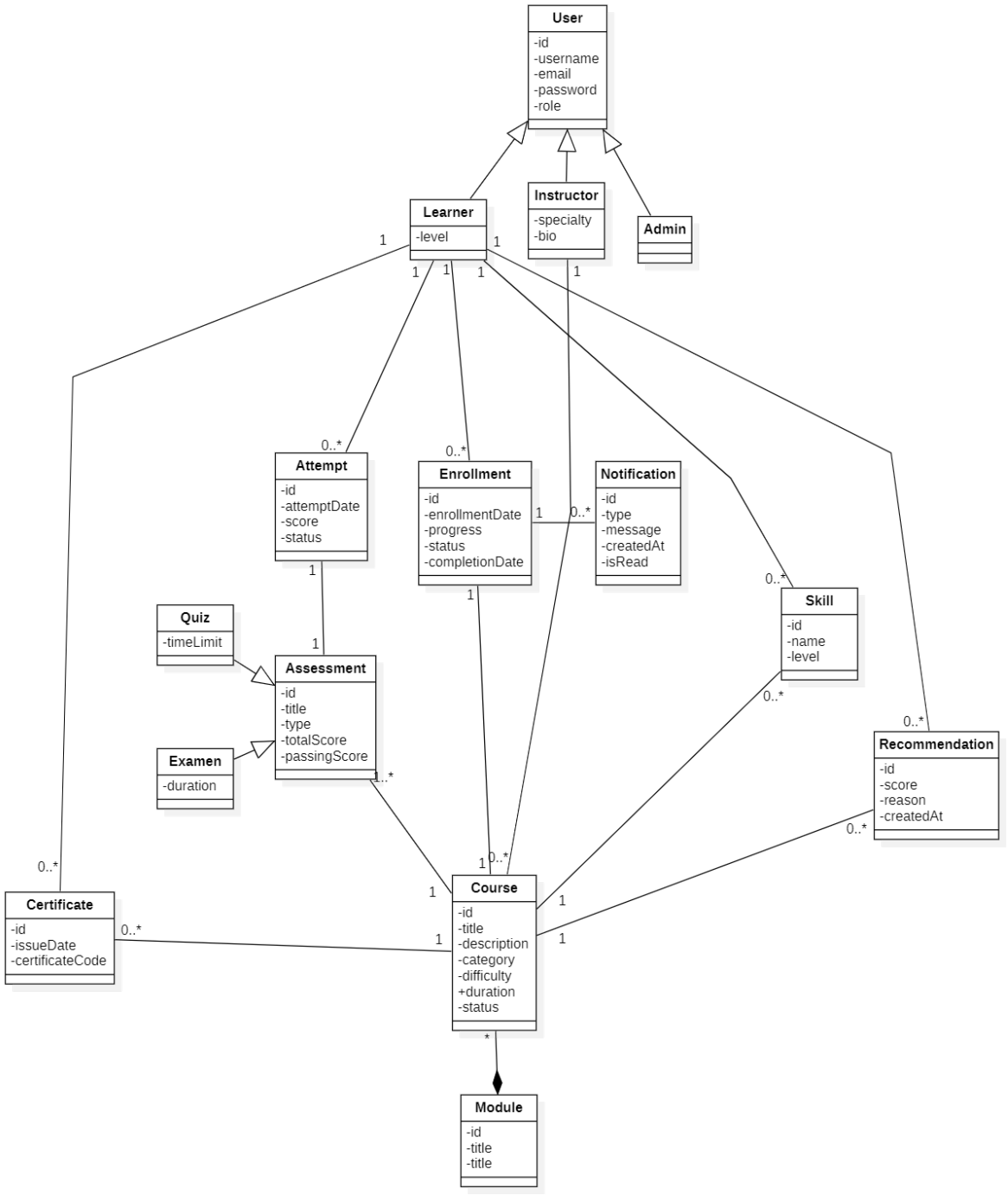

Figure 2.4: UML Class Diagram – Analysis Model of the Training Management Platform

2.5 Prototyping

Prototyping was carried out in order to validate the structure, navigation, and usability of the SkillHub Pro platform before implementation. Low-fidelity wireframes were created using Balsamiq to focus on layout, information organization, and user interaction rather than visual aesthetics. These prototypes helped refine user requirements and ensure that the interface meets the needs of learners, instructors, and administrators.

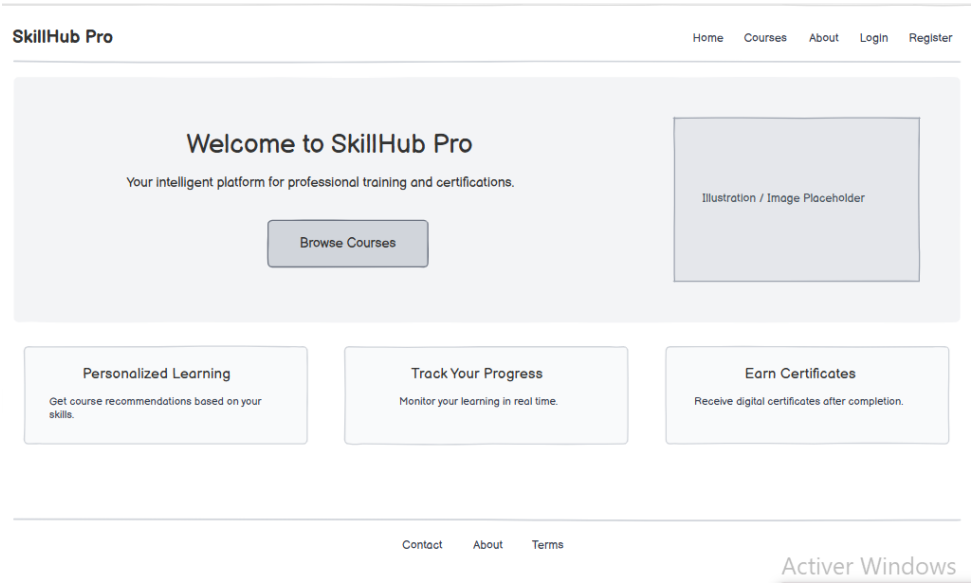

Figure 2.5: Low-Fidelity Prototype of the Home Page

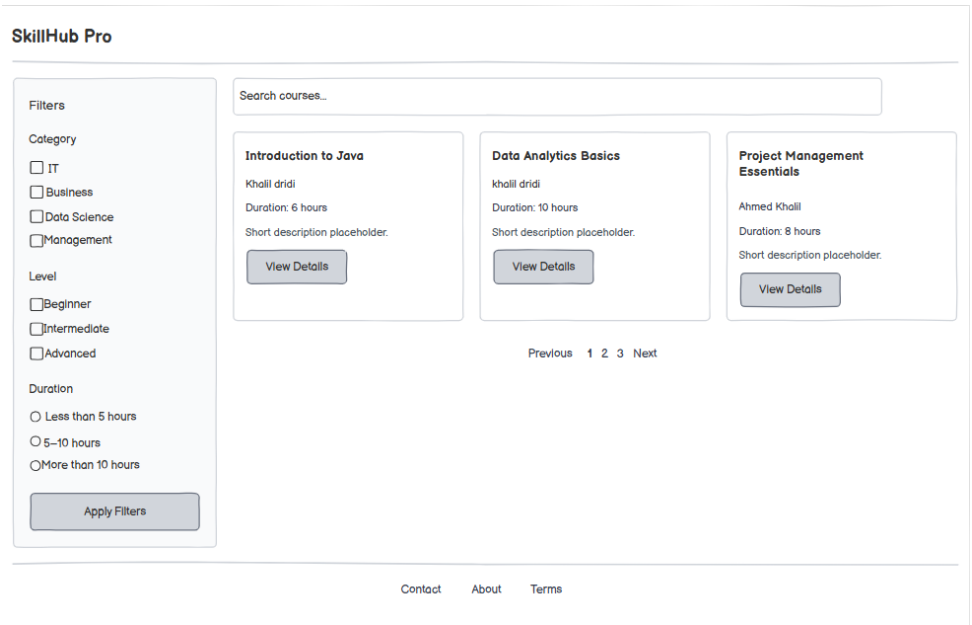

Figure 2.6: Low-Fidelity Prototype of the Course Catalog

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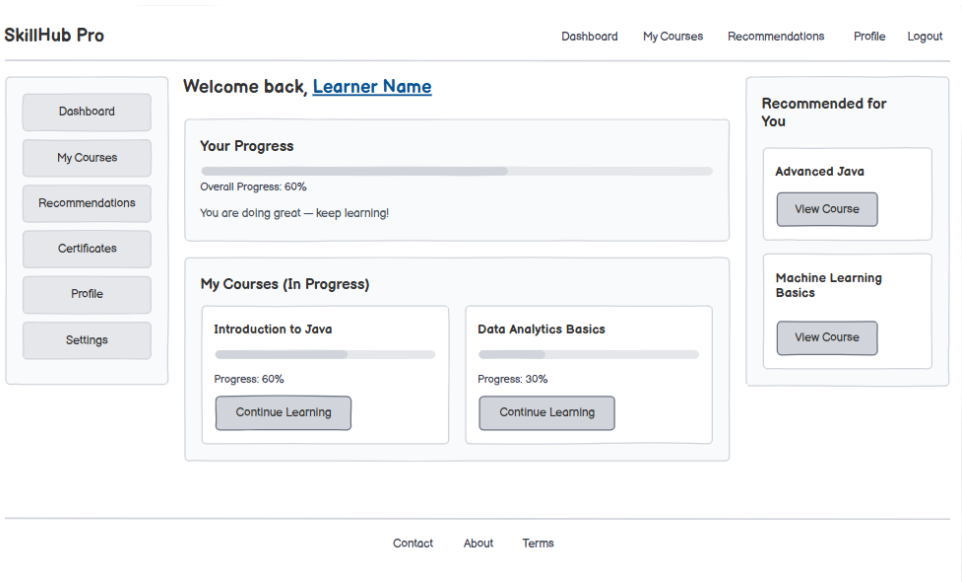

Figure 2.7: Low-Fidelity Prototype of the Learner Dashboard

2.6 Product Backlog

The Product Backlog defines and prioritizes the main user stories of the SkillHub Pro platform. Each item represents a functional or technical requirement, with priorities based on business value and estimations expressed in days.

ID	User Story	Priority	Est. (days)	Acceptance Criteria
PB1	As a visitor, I want to register and log in to the platform.	High	2	User can create an account and authenticate securely.
PB2	As a learner, I want to view and edit my profile.	High	1	Profile information can be updated and saved correctly.
PB3	As a learner, I want to browse, search, and filter courses.	High	3	Catalog displays correctly with search and filters.
PB4	As an instructor, I want to create, update, and delete courses.	High	5	Instructor can fully manage course content.
PB5	As a learner, I want to enroll in courses.	High	2	Enrollment is stored and visible in dashboard.
PB6	As a learner, I want to track my learning progress.	Medium	3	Progress indicators update in real time.
PB7	As a learner, I want to take assessments and view results.	Medium	3	Scores are calculated and stored correctly.
PB8	As a learner, I want to download certificates.	Medium	2	Certificates are generated and downloadable.

ID	User Story	Priority	Est. (days)	Acceptance Criteria
PB9	As a learner, I want to receive intelligent course recommendations.	High	5	Relevant courses are suggested based on profile and history.
PB10	As an administrator, I want to manage users and roles.	Medium	3	Roles and permissions are applied correctly.
PB11	As a training manager, I want to access analytics dashboards.	Medium	4	Dashboards display correct statistics.
PB12	As a system, I must ensure data security and GDPR compliance.	High	3	Passwords are hashed and data is encrypted.
PB13	As a learner, I want to receive notifications about my courses.	Medium	2	Notifications are sent when updates occur.
PB14	As an instructor, I want to monitor learners' progress.	Medium	3	Instructor can view learner statistics.
PB15	As an administrator, I want to suspend or delete users.	Medium	2	Admin can control user access.
PB16	As a learner, I want to reset my password.	High	1	Password recovery works via email.
PB17	As a system, I want to store data in a secure database.	High	2	Database is protected with access control.
PB18	As a learner, I want to see course details before enrolling.	High	2	Course description, duration, and skills are visible.
PB19	As an instructor, I want to upload learning materials.	High	3	Files can be uploaded and accessed by learners.
PB20	As a system, I want to expose REST APIs for integration.	Low	3	APIs are documented and functional.

Table 2.3: Product Backlog of SkillHub Pro

Conclusion

This chapter presented the planning and system architecture of SkillHub Pro. It identified the main stakeholders, defined functional and non-functional requirements, and detailed the logical and physical architecture of the platform based on the MVC and n-tier models. UML diagrams

and low-fidelity prototypes were also introduced to clarify system structure and user interaction. Together, these elements provide a solid foundation for the design and implementation phases described in the following chapters.

Appendix

.1 Examples of Existing Training Platforms

This appendix presents visual examples of existing online training platforms that were analyzed in Chapter 1. These platforms were selected because they are widely used in academic and professional contexts and illustrate different approaches to online learning, certification, and training management. The objective of this appendix is to provide visual references that support the comparative study of existing solutions.

Coursera Platform

Figure 8 illustrates the course catalog interface of Coursera, showing how courses are organized and presented to users.

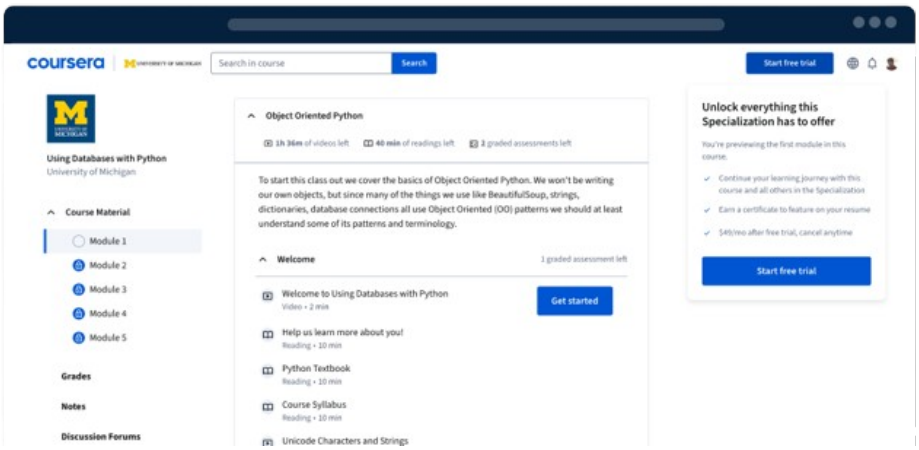

Figure 8: Coursera platform – course catalog interface

Udemy Platform

Figure 9 presents the Udemy learning dashboard, which allows learners to monitor their progress and access their enrolled courses.

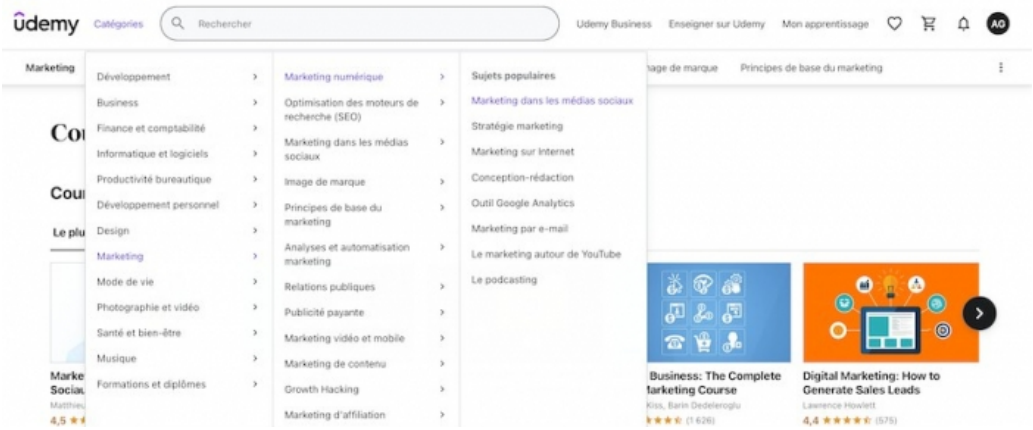

Figure 9: Udemy platform – learning dashboard interface

Moodle Platform

Figure 10 shows a typical Moodle interface, illustrating its structure for managing courses and learning activities.

Figure 10: Moodle platform – learning management system interface

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